



Start 350

About this manual

This manual describes the functionality of the starter units that are produced after 01/05-2026 with serial number ≥ 213320 .

For information on the older units, refer to the previous revision of the manual.

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1 Introduction and safety

1.1 Introduction

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Purpose of the manual

The purpose of this manual is to provide necessary information for installation, operation, and maintenance of the unit.

Read and keep the manual

Save this manual for future reference, and keep it readily available at the location of the unit.



CAUTION:

Read this manual carefully before installing and using the product. Improper use of the product can cause personal injury and damage to property, and may void the warranty.

The equipment, and its functioning, may be impaired if used in a manner not specified by the manufacturer.

Intended use



WARNING:

Operating, installing, or maintaining the unit in any way that is not covered in this manual could cause death, serious personal injury, or damage to the equipment and the surroundings. This includes any modification to the equipment or use of parts not provided by Xylem. If there is a question regarding the intended use of the equipment, please contact a Xylem representative before proceeding.

1.2 Safety terminology and symbols

About safety messages

It is extremely important that you read, understand, and follow the safety messages and regulations carefully before handling the product. They are published to help prevent these hazards:

- Personal accidents and health problems
- Damage to the product and its surroundings
- Product malfunction

Hazard levels

Hazard level		Indication
	DANGER:	A hazardous situation which, if not avoided, will result in death or serious injury
	WARNING:	A hazardous situation which, if not avoided, could result in death or serious injury
	CAUTION:	A hazardous situation which, if not avoided, could result in minor or moderate injury
NOTICE:		Notices are used when there is a risk of equipment damage or decreased performance, but not personal injury.

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Special symbols

Some hazard categories have specific symbols, as shown in the following table.

Electrical hazard		Magnetic fields hazard	
	Electrical Hazard:		CAUTION:

1.3 User safety

Introduction

All government regulations, local health and safety directives must be observed.

Prevent danger due to electricity

All danger due to electricity must be avoided. Electrical connections must always be carried out in compliance with the following:

- The standard connections shown in the product documentation that is delivered together with the product
- All international, national, state, and local regulations. For details, consult the regulations of your local electricity supplier.

For more information about requirements, see sections dealing specifically with electrical connections.

1.3.1 Power lock-out

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DANGER: Electrical Hazard

Before starting work on the unit, make sure that the unit and the control panel are isolated from the power supply and cannot be energized. This applies to the control circuit as well.



1.3.2 Qualification of personnel



WARNING: Electrical Hazard

Risk of electrical shock or burn. A certified electrician must supervise all electrical work. Comply with all local codes and regulations.

All work on the product must be carried out by certified electricians or Xylem authorized mechanics.

Xylem disclaims all responsibility for work done by untrained, unauthorized personnel.

1.4 End-of-life product disposal

Handle and dispose of all waste in compliance with local laws and regulations.

EU and UK only: Correct disposal of this product — waste electrical and electronic equipment

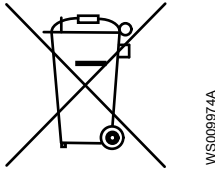
- EU: Directive 2012/19/EU on waste electrical and electronic equipment (WEEE)
- UK: SI 2013 No. 3113



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This marking on the product, accessories, or literature shows that the product should not be disposed of with other waste at the end of its working life.

EU and UK only: Correct disposal of batteries in this product



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This marking on the battery, manual, or packaging shows that the batteries in this product should not be disposed of with other waste at the end of its working life. Where marked, the chemical symbols Hg, Cd, or Pb indicate that the battery contains mercury, cadmium, or lead above the reference levels in 2006/66/EC or UK SI 2008 No. 2164. If batteries are not properly disposed of, these substances can cause harm to human health or the environment.

To protect natural resources and to promote material reuse, please separate batteries from other types of waste and recycle them through your local, free battery return system.

1.5 Spare parts



CAUTION:

Only use the manufacturer's original spare parts to replace any worn or faulty components. The use of unsuitable spare parts may cause malfunctions, damage, and injuries as well as void the warranty.

1.6 Warranty

For information about warranty, see the sales contract.

1.7 Support

Xylem only supports products that have been tested and approved. Xylem does not support unapproved equipment.

2 Product description

2.1 Product design

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START 350 is a manual starter. The pump is started with the handles, inset into the front.

Automatic operation of the pump, for example with level control equipment, is not possible.

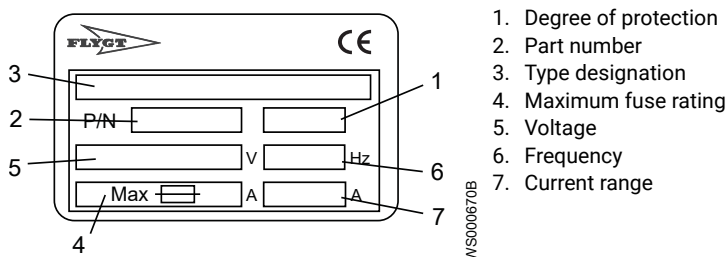
Overload protection

The built-in motor protection switch, which has a thermomagnetic tripping function, protects both the pump and its cable against overload and short circuit.

Current ranges

It is available for all current ranges from 1.0 to 25 A, and with built-in CEE plug for simple connection to the power supply.

2.2 The data plate



2.3 Technical data

Table 1: Electrical characteristics

Parameter	Value
Supply voltage *	3 x 380–415 V 3 x 440–500 V
Rated current *	25 A maximum
Frequency	50–60 Hz
Maximum power supply fuse rating	See Maximum fuse rating on page 7.
Mains connection	CEE 16 A: 3P+N+PE CEE 32 A: 3P+N+PE
Pump connection	Cable gland, M32, 8–18 mm
* See the technical data that is supplied with the starter.	

Table 2: Physical characteristics

Parameter	Value
Material	Polycarbonate
Dimensions (H x W x D)	300 x 90 x 87 mm (11.8 x 3.5 x 3.4 in)
Weight	Approximately 0.9 kg (2 lb)

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Table 3: Features

Feature	Value
Thermal trip	Yes
Magnetic trip	Yes
Temperature compensation	Yes, automatic between -25°C to +50°C (-13°F to 122°F)
Motor temperature monitoring	No
Phase sequence indication	Yes
Degree of protection	IP 45
Ambient temperature	-25° to +40°C (-13°F to 104°F)

Table 4: Maximum connection area

Conductor	Number	Minimum	Maximum
Solid conductor	1 x	1	10 mm ²
	2 x	1	4 mm ²
Multiple-strand conductor	1 x	1	6 mm ²
	2 x	—	—

2.4 Maximum fuse rating

Current range	230 V	400 V	500 V
1.0-1.6 A	No back-up fuse needed.	No back-up fuse needed.	25 A
1.6-2.5 A			25 A
2.5-4.0 A			35 A
4.0-6.3 A			35 A
6.3-10.0 A			40 A
8.0-12.0 A		50 A	50 A
10.0-16.0 A	63 A	63 A	63 A
16.0-20.0 A	80 A	80 A	80 A
20.0-25.0 A	100 A	100 A	100 A

3 Installation

3.1 Facility requirements

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Exposure to water or physical damage

The starter has a high degree of protection against moisture and dirt, but it should always be installed so that it will not be unnecessarily exposed to water or the risk of external physical damage.

Explosive zones

Do not install the starter equipment in an explosive zone.

NOTICE:

Do not use this unit in environments that may contain flammable/explosive or chemically aggressive gases or powders.

3.2 Make the electrical connections



DANGER: Electrical Hazard

Before starting work on the unit, make sure that the unit and the control panel are isolated from the power supply and cannot be energized. This applies to the control circuit as well.



WARNING: Electrical Hazard

There is a risk of electrical shock or explosion if the electrical connections are not correctly carried out, or if there is fault or damage on the product. Visually inspect equipment for damaged cables, cracked casings or other signs of damage. Make sure that electrical connections have been correctly made.



WARNING: Electrical Hazard

Risk of electrical shock or burn. A certified electrician must supervise all electrical work. Comply with all local codes and regulations.

**DANGER: Electrical Hazard**

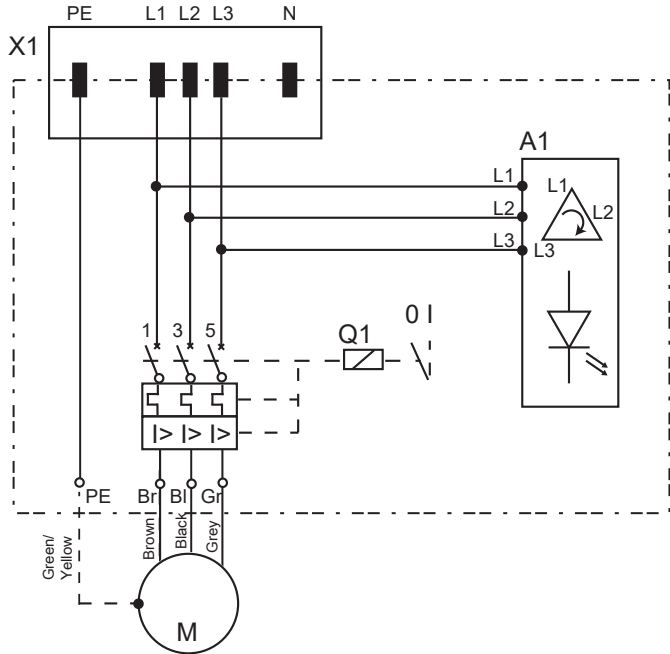
All electrical equipment must be grounded (earthed). Test the ground (earth) lead to verify that it is connected correctly and that the path to ground is continuous.

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1. Check that the starter rating agrees with the pump rating and with the current, voltage and frequency of the power supply system.
2. Check that the fuse in the power supply agrees with the maximum permissible fuse rating.
See [Maximum fuse rating](#) on page 7.
3. Make the electrical connections in accordance with the connection diagram.
To ensure that the pump will have the correct direction of rotation, it is important to follow both the phase sequence and color sequence of the supply cable and pump cable.
4. Set the current setting of the motor protection switch to the rated current of the motor.
The breakers are automatic temperature compensated between -20° – +50°C.
5. Check that all connections have been securely tightened. Give particular attention to the protective conductors.

3.3 Wiring diagram

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Terminal	Description
X1	CEE plug
Q1	Motor protection breaker with shunt trip coil
A1	Phase sequence indicator
M	Pump motor

3.4 External connections

Table 5: Motor Protection Breaker

Terminal Markings	Pump Cable Wires
PE	Green/yellow
Br	Brown
Bl	Black
Gr	Grey

3.5 Commissioning

1. Check that the starter is closed before the power supply is turned on.
2. Switch on the main power supply or fit the main fuses.
3. Check that the phase sequence of the power supply is correct. If the red indicating lamp is lit, then the phase sequence is incorrect.

If the power supply phase sequence is incorrect, then change it by following this procedure:

- a) Isolate the unit from the power supply.
- b) Change the phase sequence:

Condition	Action
Units with built-in CEE plug	Transpose two phase pins in the CEE connector. This applies only to 5-pole CEE connectors.
Starters without transposable phase pins	Transpose two of the phase conductors in the incoming power supply.

4. Check the direction of rotation of the pump, by following the directions in the Installation, Operation, and Maintenance Manual for the pump.

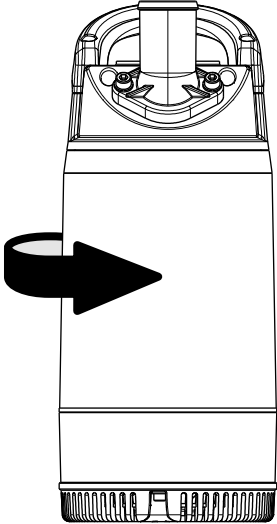


CAUTION: Crush Hazard

The starting jerk can be powerful. Make sure nobody is close to the unit when it is started.

The pump will jerk in an anti-clockwise direction if the impeller is rotating correctly.

Figure 1: Start reaction



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4 Troubleshooting

4.1 Precautions

Before starting work, make sure that the safety instructions have been read and understood.



DANGER: Electrical Hazard

Troubleshooting a live control panel exposes personnel to hazardous voltages. Electrical troubleshooting must be done by a qualified electrician.



DANGER: Electrical Hazard

Before starting work on the unit, make sure that the unit and the control panel are isolated from the power supply and cannot be energized. This applies to the control circuit as well.



4.2 Common problems

Fault	Possible Reason	Action
The red lamp indicating incorrect phase sequence is lit.	Incorrect phase sequence	Change the phase sequence of the incoming power supply.
The pump fails to start	No power supply to the pump	Check the main fuses. Check that the main switch is closed.
	The motor protection has tripped.	Let the motor cool down, and then restart the pump. If the switch trips again, then see possible reason "Overcurrent".

Fault	Possible Reason	Action
The pump rotates in the wrong direction	Either of the following: - The motor cable phase conductors are transposed. - The phase sequence in the power supply is wrong; power supply is not connected in accordance with the connection diagram.	Check that the connection agrees with the connection diagram and correct if necessary.
The pump starts but runs unsatisfactorily	Low voltage	Cables are too long or underdimensioned.
	Open circuit in one of the phases	Check the main fuses.
The motor protection breaker trips.	Incorrect setting range	Check on the rating plates that the rated current ranges of the pump and starter agree.
	Incorrect current setting	Adjust to the rated current of the motor.
	Low voltage	Cables are too long or underdimensioned.
	Loss of one phase	Check the main fuses.
	Overcurrent	The impeller is clogged. - Let the motor cool down, and then reset the starter. - To clear the impeller of any blockage, make a few short starting attempts. If the starter initiates an alarm again, then the pump must be taken out of operation for service.

4.3 Spare parts

Spares are normally not necessary. However, abnormal conditions can result in parts sustaining damage and requiring replacement. Examples of such conditions include:

- Incorrect handling or installation
- Wrongly dimensioned fuses
- Undersized cables
- Low voltage

The following are examples of components that normally can sustain damage under these conditions:

- Motor protection breaker
- Auxiliary relays

- Contactors
- Capacitors
- Start relays

If spare parts are required, then exact component data can be obtained from the electrical documentation that is supplied with the starter. The parts can be ordered through the nearest service office.

If the documentation is not available and spare parts are needed, then the starter part number or the serial number on the data plate must be stated.

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